Computational Fluid Dynamics
EN 530.767
HW 3

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1. Objective

Explore a Gaussian vortex in a freestream the convection of which can be modeled by viscous Burger's equation:

$$u_t + uu_x + vu_y = \mu(u_{xx} + u_{yy})$$
$$v_t + uv_x + vv_y = \mu(v_{xx} + v_{yy})$$

The velocity field of this vortex is

$$u = 1 - V_t(y - y_0)\exp\left(\frac{1-(r/r_0)^2}{2}\right)$$
$$v = V_t(x - x_0)\exp\left(\frac{1-(r/r_0)^2}{2}\right)$$

where $V_t = 0.25, x_0 = 0.5;\ y_0 = 0.5, r = \sqrt{(x-x_0)^2 + (y-y_0)^2}$ and $r_0 = 0.1$.

Discretize the above equation using a central difference scheme in space, Forward Euler for the convection term and backward Euler for the diffusion term.

2. Methodology

2.1 Numerical schemes

To solve this 2-dimensional Burger's equation, the space is discretized using a central difference scheme, and the time is discretized by forward Euler method for the convection term and the backward Euler method for the diffusion term. The discretized scheme for the equation about u can be written as:

$$\frac{u_{i,j}^{n+1} - u_{i,j}^{n}}{\Delta t} + u_{i,j}^{n}\frac{u_{i,j+1}^{n} - u_{i,j-1}^{n}}{2\Delta x} + v_{i,j}^{n}\frac{u_{i+1,j}^{n} - u_{i-1,j}^{n}}{2\Delta y}$$
$$= \mu\frac{u_{i,j+1}^{n+1} - 2u_{i,j}^{n+1} + u_{i,j-1}^{n+1}}{\Delta x^2} + \mu\frac{u_{i+1,j}^{n+1} - 2u_{i,j}^{n+1} + u_{i-1,j}^{n+1}}{\Delta y^2}$$

Using Gauss-Seidel scheme, the discretized scheme can be written as:

$$\left(1+2\frac{\mu\Delta t}{\Delta x^2}+2\frac{\mu\Delta t}{\Delta y^2}\right)u_{i,j}^{n+1}$$
$$= -u_{i,j}^{n}\frac{\Delta t}{2\Delta x}\left(u_{i,j+1}^{n}-u_{i,j-1}^{n}\right)-v_{i,j}^{n}\frac{\Delta t}{2\Delta y}\left(u_{i+1,j}^{n}-u_{i-1,j}^{n}\right)$$
$$+\frac{\mu\Delta t}{\Delta x^2}\left(u_{i,j+1}^{n+1}+u_{i,j-1}^{n+1}\right)+\frac{\mu\Delta t}{\Delta y^2}\left(u_{i+1,j}^{n+1}+u_{i-1,j}^{n+1}\right)$$

The discretized scheme for the equation about v is similar.

2.2 Numerical set up

The geometry and boundary conditions are shown in Figure 1.

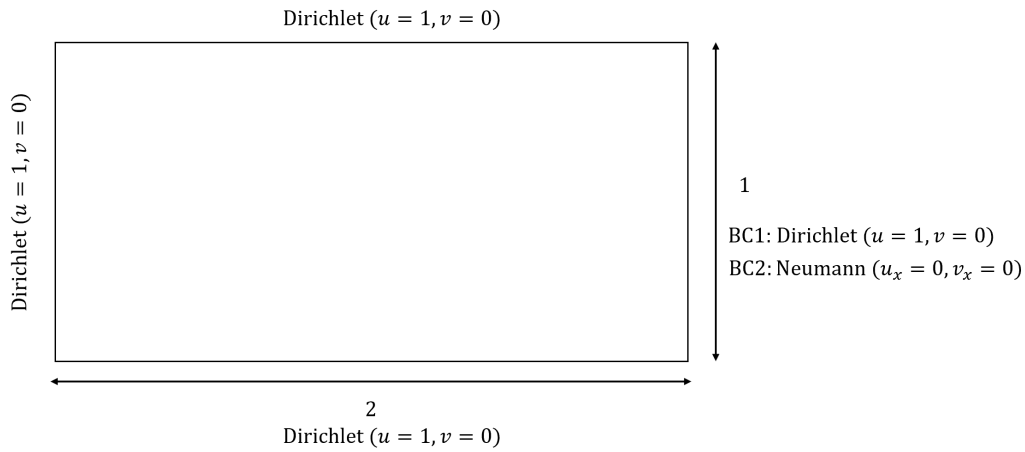

Figure 1 Geometry and boundary conditions of set up

The domain size is $L_x \times L_y = 2 \times 1$.The Dirichlet boundary condition is applied to the left, top and bottom boundaries with $u = 1, v = 0$. Two different boundary conditions are examined on the right boundary. The first one is the Dirichlet boundary condition with $u = 1, v = 0$, and the second one is the Neumann boundary condition with $u_x = 0, v_x = 0$.

2.3 Parameters defined

Given the domain geometry and the boundary conditions, this study examines the convection of the vortex from time $t = 0{\sim}2.0$ with $\mu = 0.01$ and $\mu = 0.001$. The effect of grid resolution has also been examined, by employing grids with spacings of $(\Delta x, \Delta y) = 0.02, 0.01, 0.005$.

3. Results and Discussions

3.1 Vorticity contours

The vorticity contours at time intervals $t = 0, 0.5, 1.0, 1.5, 2.0$ with Dirichlet and Neumann boundary conditions with different μ and different $(\Delta x, \Delta y)$ are plotted in Figure 2-7. In these figures, we can observe how the vortices evolve with time, as well as how the shapes and amplitudes of the vortices change affected by modified wave numbers.

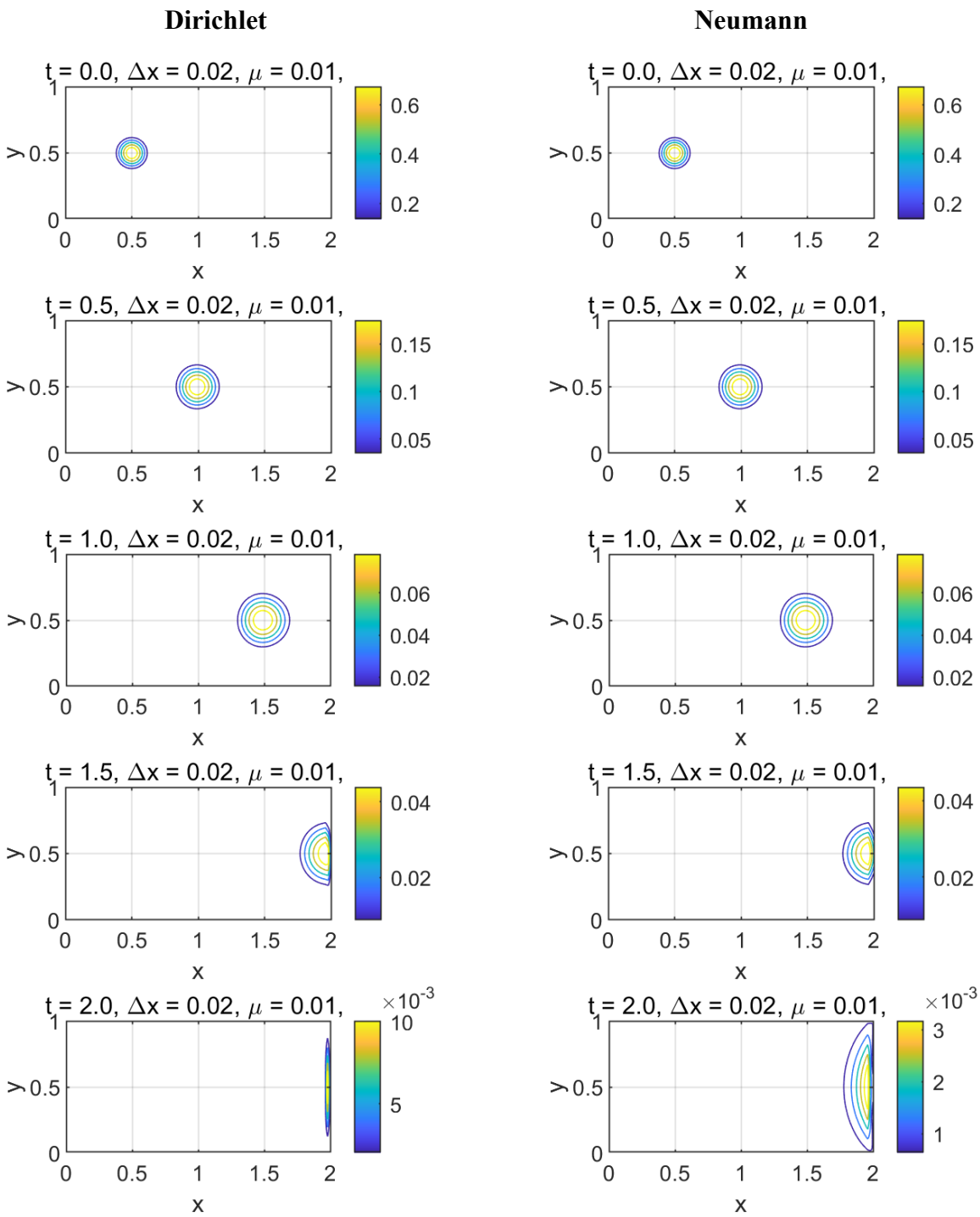

Figure 2 Vorticity Contours with two boundary conditions at $\Delta x = 0.02$, $\mu = 0.01$

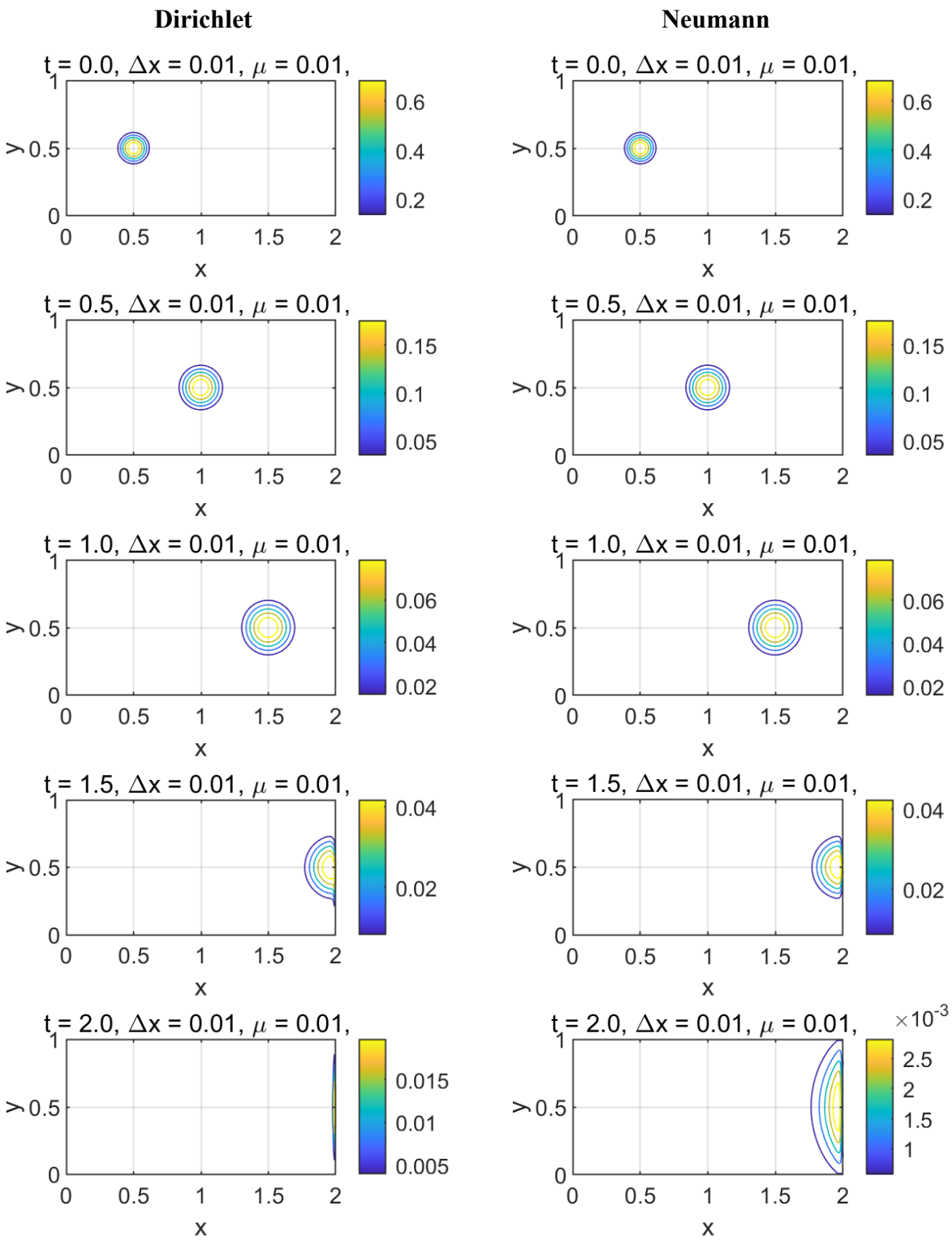

Figure 3 Vorticity Contours with two boundary conditions at $\Delta x = 0.01$, $\mu = 0.01$

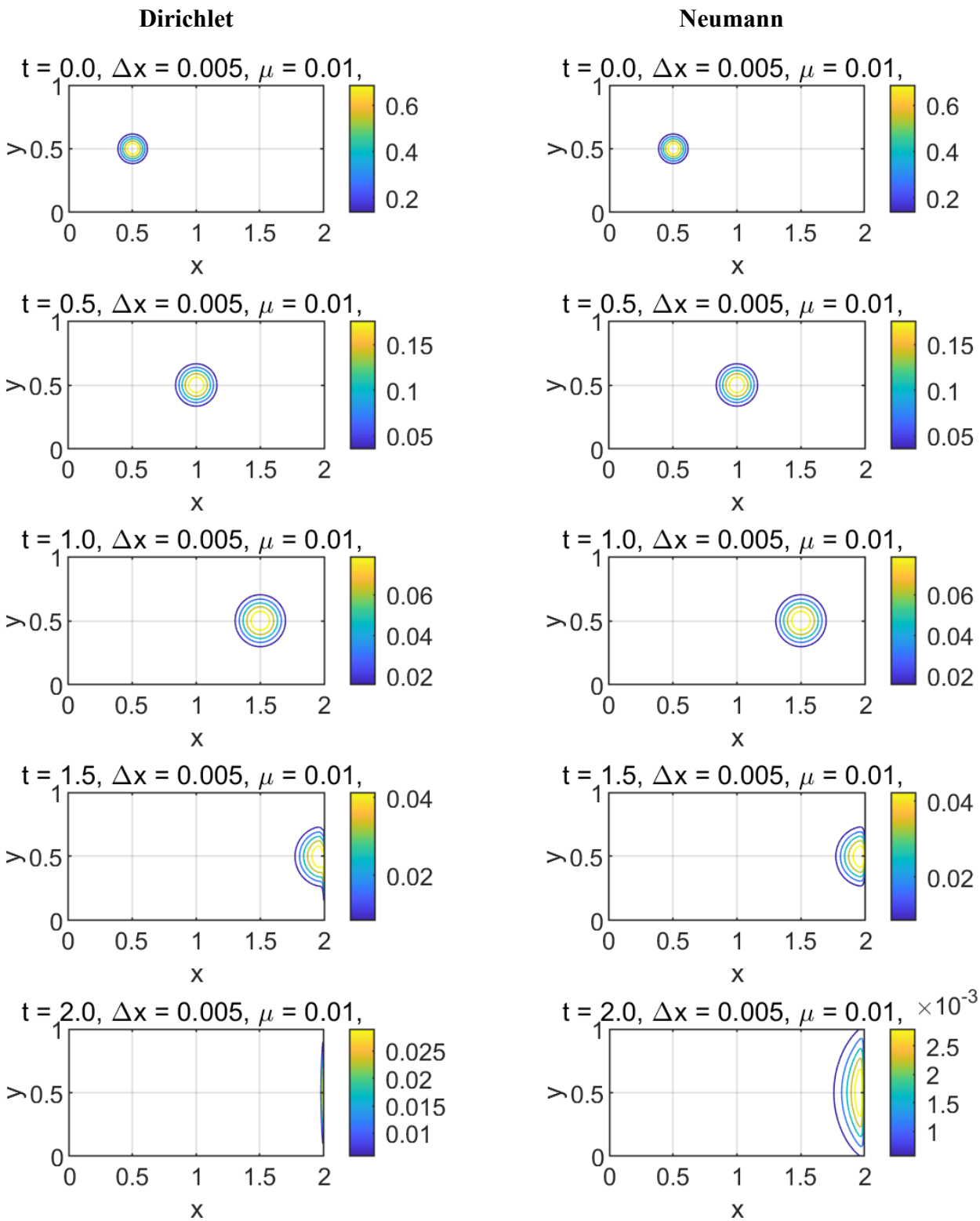

Figure 4 Vorticity Contours with two boundary conditions at $\Delta x = 0.005$, $\mu = 0.01$

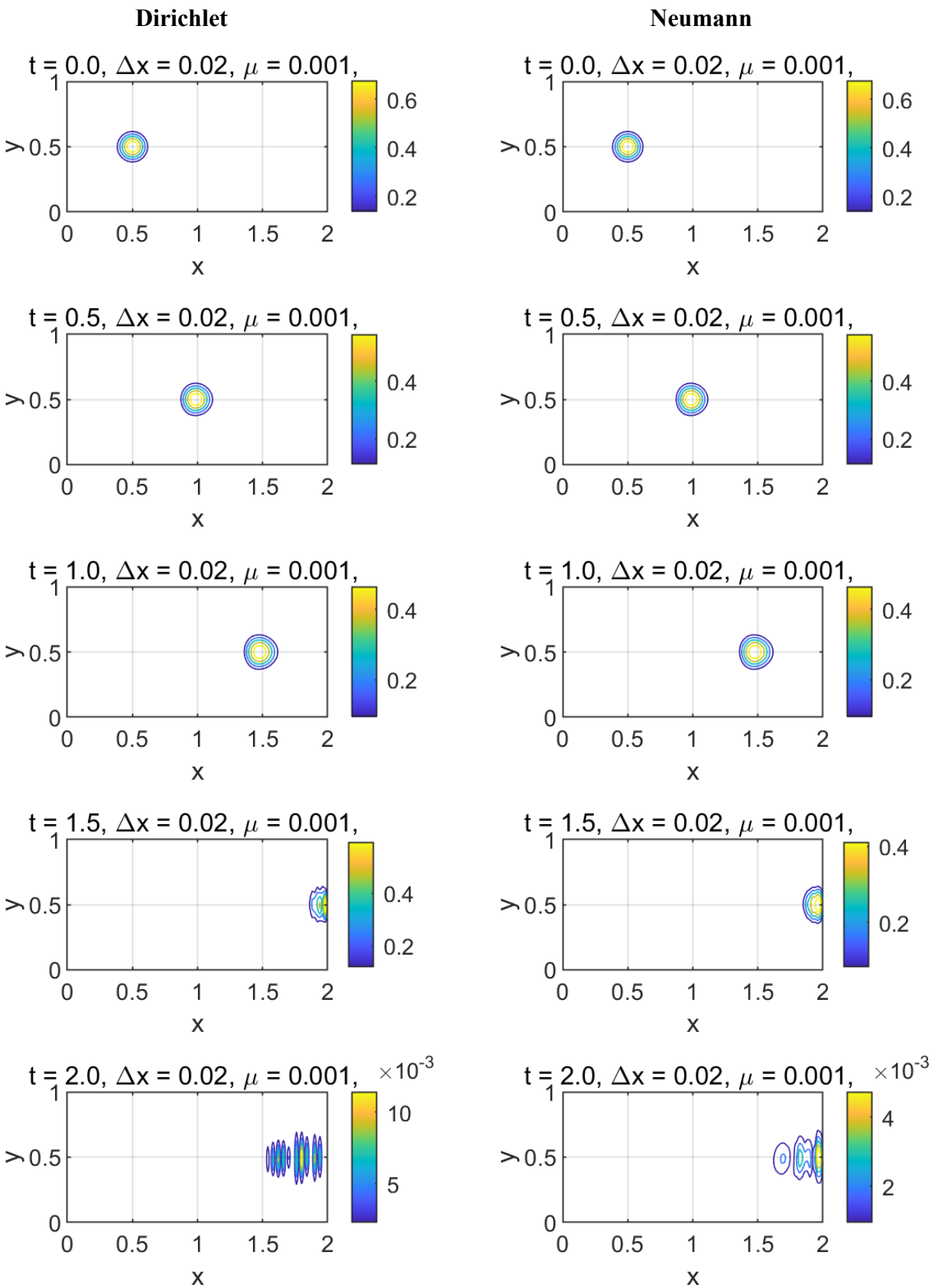

Figure 5 Vorticity Contours with two boundary conditions at $\Delta x = 0.02$, $\mu = 0.001$

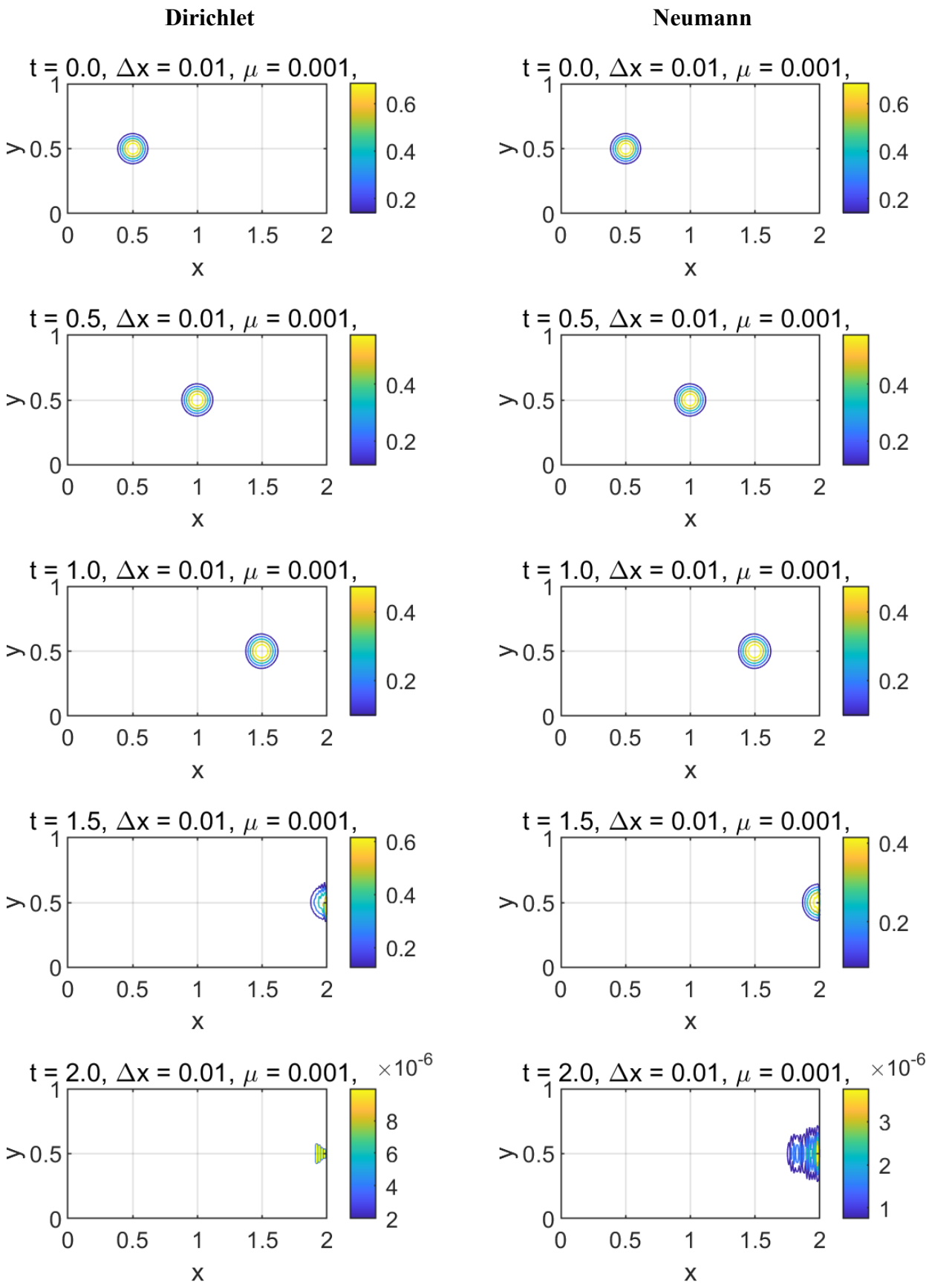

Figure 6 Vorticity Contours with two boundary conditions at $\Delta x = 0.01$, $\mu = 0.001$

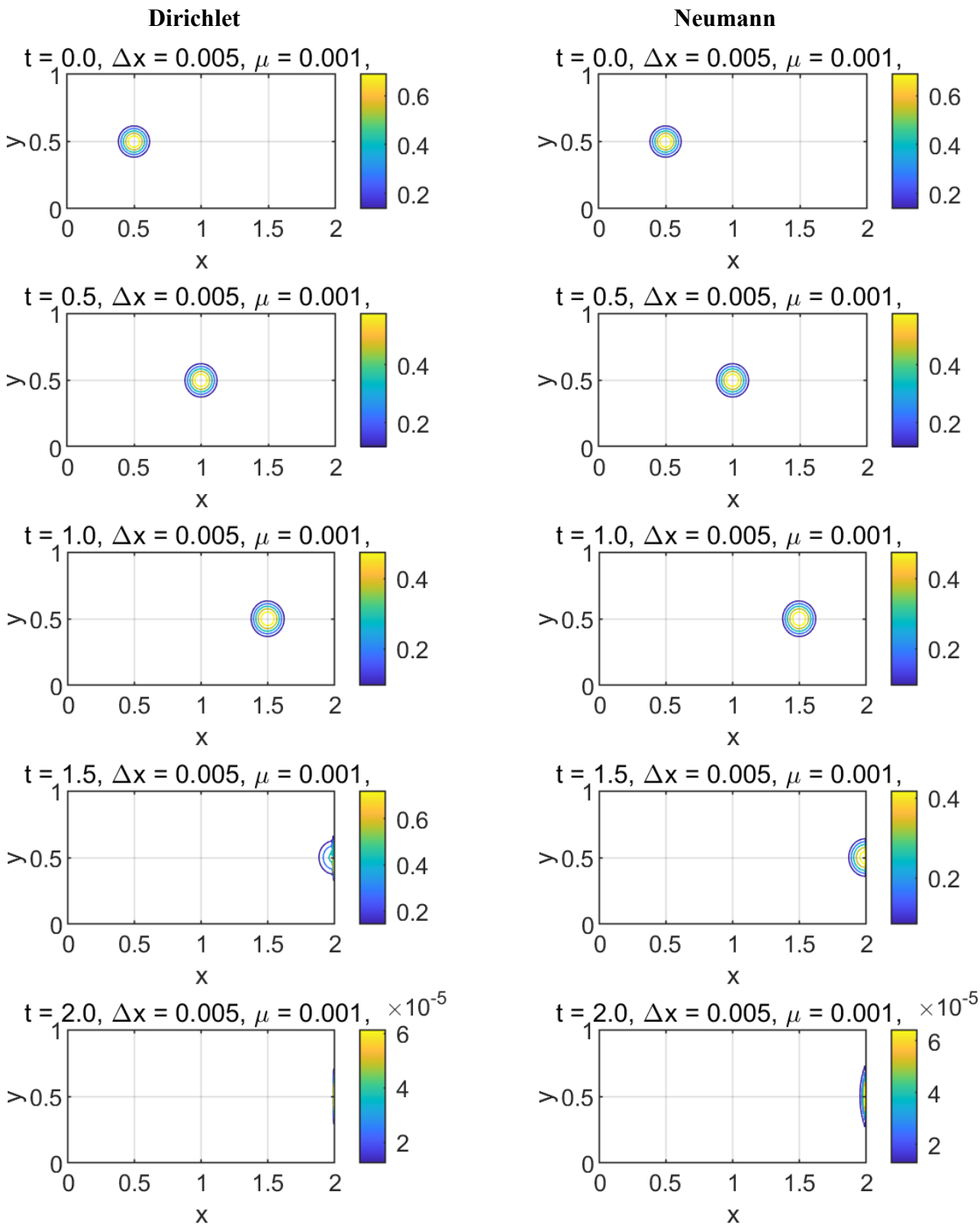

Figure 7 Vorticity Contours with two boundary conditions at $\Delta x = 0.005$, $\mu = 0.001$

The vortex in the flow field moved towards the positive x-direction at a convective speed of 1 unit length per unit time, corresponding to the boundary velocity where $u = 1$ and $v = 0$. Over time, the vortex experiences an increase in size, accompanied

by a decrease in amplitude. This phenomenon is attributed to diffusion influenced by the viscosity μ. The vortex diffuses outward from a central point, spreading to the surrounding areas. Comparative analysis of Figures 2 and 5, 3 and 6, and 4 and 7, all under identical boundary conditions and grid resolutions with different viscosity, reveals a key observation: the vortex with a higher viscosity ($\mu = 0.01$) expands faster than that with a slower viscosity ($\mu = 0.001$), and the vorticity magnitude decreases sharper. This implies that a flow field with greater viscosity experiences faster diffusion, which is consistent with theoretical expectations. In addition, for $\mu = 0.001$, there are some apparent oscillations because of the boundary when the grid resolution is relatively large (e.g. $\Delta x = 0.01, 0.02$), while there is no oscillation when $\mu = 0.01$, which is more stable because of the larger viscosity.

3.2 Modified wave numbers

The derivation of modified wave numbers for the convection terms is shown in Figure 8, and the corresponding modified wave number curves are plotted in Figure 9.

$u_i = \hat{u} e^{ikx}$

For the central difference scheme of the convection term

$u_t = -uu_x = -iku_i^2$

This k is the wave number of exact solution

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = -u_i^n \frac{u_{i+1}^n - u_{i-1}^n}{2\Delta x} = -ik' u_i^2$$

$$\Rightarrow -u_i^2 \frac{e^{ik\Delta x} - e^{-ik\Delta x}}{2\Delta x} = -ik' u_i^2$$

$$\Rightarrow k' = \frac{\sin(k\Delta x)}{\Delta x}$$

For the central difference scheme of the diffusion term

$u_t = \mu u_{xx} = -i\mu k u_i$

This k is the wave number of exact solution

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = \mu \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2} = -i\mu k' u_i$$

$$\Rightarrow u_i \frac{e^{ik\Delta x} - 2 + e^{-ik\Delta x}}{\Delta x^2} = u_i \frac{2\cos(k\Delta x) - 2}{\Delta x^2} = -ik' u_i$$

$$\Rightarrow -ik' = \frac{2\cos(k\Delta x) - 2}{\Delta x^2}$$

$$\Rightarrow k' = -i \frac{2 - 2\cos(k\Delta x)}{\Delta x^2}$$

This is similar for y direction and v velocity.

Figure 8 Derivation of modified wave numbers of convection and diffusion terms

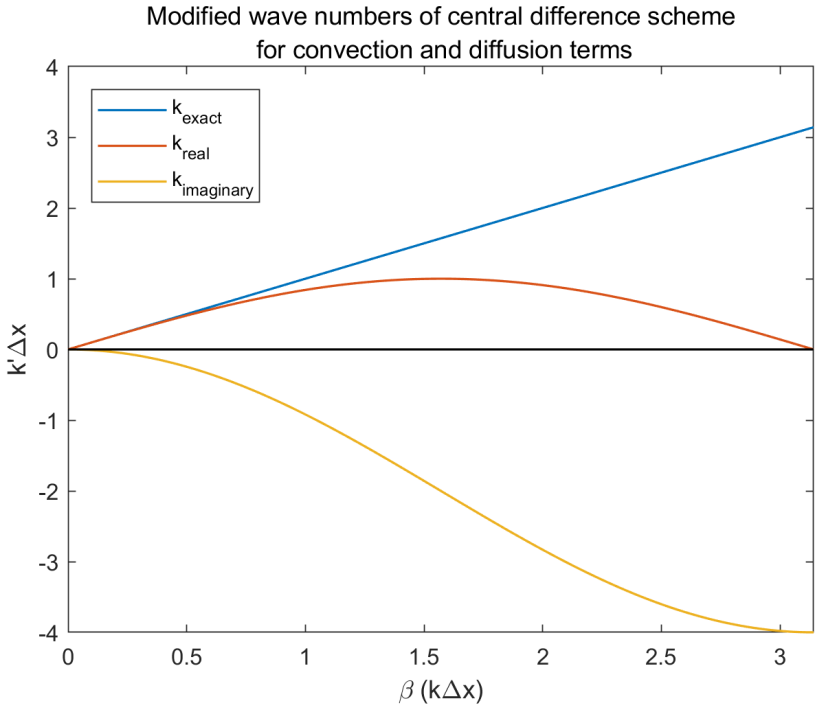

Figure 9 Modified wave numbers of central difference scheme for convection and diffusion terms

The modified wave number of the convection term constitutes the real part of the modified wave number of the equation, which represents the convective speed of the solution, affecting the shape of the vortex. On the other hand, the modified wave number of the diffusion term constitutes the imaginary part of the modified wave number of the equation, which represents the diffusive speed of the solution, affecting the amplitude of the vorticity of the vortex. The influence of the modified wave number can be observed in the corresponding vorticity contour patterns.

According to Figure 9, as the real part of the modified wave number increases, the discrepancy between the exact wave number (k_{exact}) and the modified wave number (k') grows, leading to more significant vortex deformation and a greater deviation from the original circular shape. This effect is evident in Figure 5, which exhibits that with $\mu = 0.001$ and $\Delta x = 0.02$, the vortex shows noticeable deformation over time. However, when $\mu = 0.001$ and $\Delta x = 0.005$, the deformation is less pronounced due to the denser grid, resulting in k' being closer to k_{exact}. Additionally, for $\mu = 0.01$, vortex deformation is negligible because the flow field with higher viscosity (μ) is less impacted by the effects of the modified wave number.

Regarding the imaginary part of the modified wave number, according to the tendency of the orange line in Figure 9, as Δx increases, the imaginary k' decreases, which indicates that the diffusion of the vortex becomes faster. This phenomenon can be observed by calculating the maximal vorticity magnitude for different grid resolution when $\mu = 0.001$. Table 1 displays the maximal vorticity magnitudes at various times and grid resolutions when μ=0.001. Notably, at T=2.0, the vortex nearly

exits the computation domain, leading to extremely small magnitudes; therefore, the magnitudes at that time are not included in the table. The table reveals a gradual decrease in the maximal vortex magnitude from $\Delta x = 0.005$ to $\Delta x = 0.01$ and then to $\Delta x = 0.02$, indicating more rapid decay at coarser grid resolutions. This trend is corroborated by the imaginary part curve in Figure 9. The numerical results of vorticity align with the expectations from the modified wave number curves.

Maximal vorticity magnitudes at different times and grid resolutions when $\mu = 0.001$					
T	0	0.5	1.0	1.5	
Dirichlet	0.8080	0.6747	0.5586	0.7149	dx = 0.02
Neumann	0.8080	0.6747	0.5586	0.4935	
Dirichlet	0.8202	0.6892	0.5696	0.7389	dx = 0.01
Neumann	0.8202	0.6892	0.5696	0.4979	
Dirichlet	0.8233	0.6919	0.5713	0.8640	dx = 0.005
Neumann	0.8233	0.6919	0.5713	0.5023	

Table 1 The maximal vorticity magnitudes at different times and grid resolutions when $\mu = 0.001$

3.3 Effect of outflow boundary conditions

The Dirichlet and Neumann boundary conditions have different effects on the inside flow and vortex. This can be characterized by extending the L_x to 3, where the vortex will not go outside of the computation domain at $t = 1.5$. The result after extending the domain is shown in Figure 10.

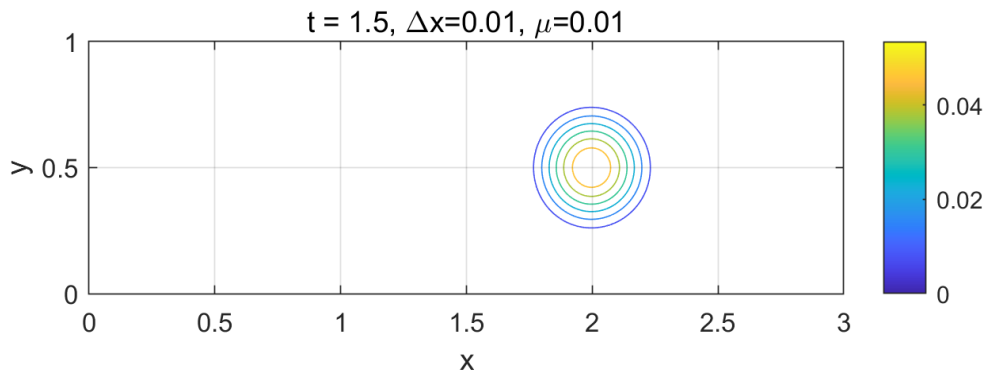

Figure 10 Vorticity Contour at $t = 1.5$ after extending to L_x to 3

Comparing the results from the original domain with Dirichlet and Neumann boundary conditions, as depicted in Figure 11 and Figure 12 respectively, it is observed that the vortex under Neumann boundary conditions resembles a halved version of the vortex in the extended domain, showing more similarity to the extended domain's vortex than the one under Dirichlet conditions. The vortex under Dirichlet conditions exhibits additional oscillations. This difference arises because Dirichlet boundary conditions artificially impose a fixed boundary, whereas Neumann boundary conditions interact with the internal flow dynamics. Consequently, the Neumann boundary condition results in less numerical disturbance to the vortex and yields smoother outcomes.

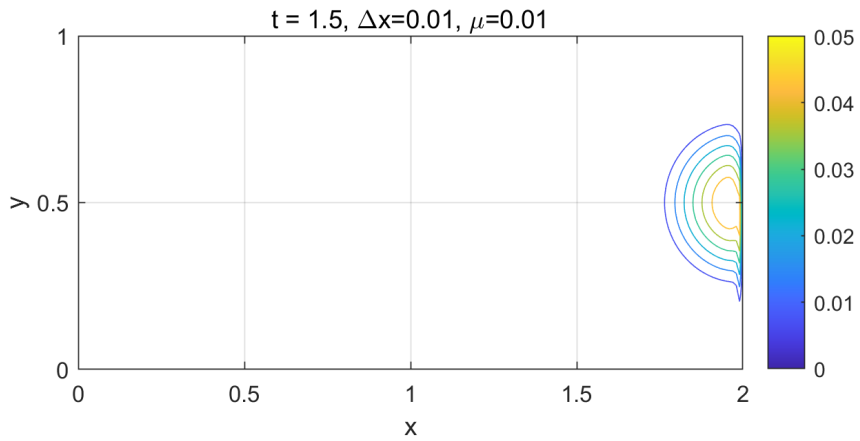

Figure 11 Vorticity Contour at $t = 1.5$ with Dirichlet boundary condition

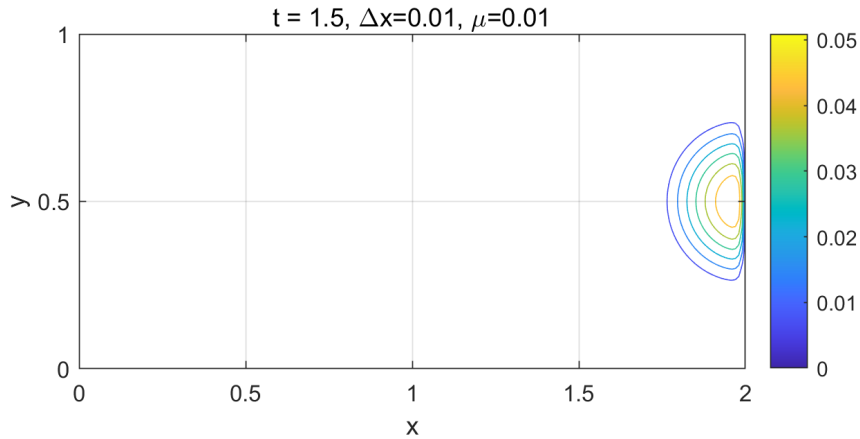

Figure 12 Vorticity Contour at $t = 1.5$ with Neumann boundary condition

4. Conclusion

In conclusion, this study explores the dynamics of a Gaussian vortex in a freestream as modeled by the viscous Burger's equation, employing numerical schemes that discretize space and time with various methods. The analysis has studied how the vortex evolves with time and how the shape and diffusion of vortex are affected by the modified wave numbers of numerical schemes. The study demonstrates that finer grid resolution results in slower diffusion and milder distortion of vortex, aligning with the expectations from the modified wave number curves. This study also explores the impact of boundary conditions on the vortex behavior, showing that the Neumann boundary condition provided a less disturbed representation of the vortex movement at the boundary compared to the Dirichlet condition. This study provides insightful observations to understanding viscous Burger's equation and Gaussian vortex behavior, enhancing our comprehension of numerical schemes and vortex dynamics.

5. Appendix

MATLAB source code:

```
%% Vorticity contours
% Plot individually
clear; clc;
mu = 0.01;
dx = 0.01;
dt = 0.001;
domain_size = 1;
k=2;
T = 1.5;
bc = 2;
[u,v] = gauss(mu, dt, dx, T, bc, k);
w = vorticity(u,v,dx,k);
w(w<1e-5)=0;
[X, Y] = meshgrid(linspace(0,k*domain_size,size(w, 2)), linspace(0,1,size(w,
1)));
maxValue = max(max(w));
minValue = min(min(w));
levels = linspace(minValue, maxValue, 8);

figure;
contour(X, Y, w, levels);
cd = colorbar;
axis equal
xticks(0:0.5:6);
yticks(0:0.5:3);
grid on;
caxis([minValue maxValue]);

title([sprintf('t = %.1f, ',T), '\Deltax = ', sprintf('%.3f, ',dx), '\mu =
', sprintf('%.3f, ',mu)]);
xlabel('x');
ylabel('y');

%% Plot together
clear; clc;
mu = 0.001;
dx = 0.02;
dt = 0.001;
k=2;
domain_size = 1;
```

```
for i = 1 : 5
    for j = 1 : 2
        bc = j;
        T = 0.5 * (i - 1);
        subplot(5,2, (i-1)*2 + j)
        [u,v] = gauss(mu, dt, dx, T, bc, k);
        w = vorticity(u,v,dx,k);
        [X, Y] = meshgrid(linspace(0,2*domain_size,size(w, 2)),
linspace(0,domain_size,size(w, 1)));
        maxValue = max(max(w));
        minValue = min(min(w));
        levels = linspace(minValue, maxValue, 10);
        caxis([minValue maxValue]);
        contour(X, Y, w, levels);
        colorbar;
        axis equal
        xticks(0:0.5:6);
        yticks(0:0.5:3);
        grid on;
        title([sprintf('t = %.1f, ',T), '\Deltax = ', sprintf('%.3f, ',dx),
'\mu = ', sprintf('%.3f, ',mu)]);
        xlabel('x');
        ylabel('y');
    end
end

%% Modified wave numbers
beta = linspace(0, pi, 100);
k_exact = beta;
ki = sin(beta);
kr = -(2-2*cos(beta));
figure;
plot(beta,k_exact,'linewidth',1)
hold on
plot(beta,ki,'linewidth',1)
hold on
plot(beta,kr,'linewidth',1)

x = linspace(0, pi, 100);
y = 0.*x;
hold on
plot(x,y,'k-','linewidth',1)

xlim([0 pi])
```

```
xlabel('\beta (k\Deltax)')
ylabel('k''\Deltax')
title('Modified wave numbers of central difference scheme',' for convection
and diffusion terms');
legend('k_{exact}','k_{real}','k_{imaginary}')

%% Functions
function [u,v] =gauss(mu, dt, dx, T, bc, k)
N = 1/dx;
dy = dx;
x = (0:k*N).*dx;
y = (0:N).*dy;
Vt = 0.25;
x0 = 0.5;
y0 = 0.5;
r0 = 0.1;
R = mu*dt/dx^2;
c = dt/(dx*2);

for i = 1:N+1
    for j = 1:k*N+1
        r(i,j) = sqrt((x(j)-x0)^2+(y(i)-y0)^2);
        u_init(i,j) = 1-Vt*(y(i)-y0)*exp((1-(r(i,j)/r0)^2)/2);
        v_init(i,j) = Vt*(x(j)-x0)*exp((1-(r(i,j)/r0)^2)/2);
    end
end

if bc == 1
    u_init(1,1:k*N+1) = 1;
    u_init(N+1,1:k*N+1) = 1;
    u_init(1:N+1,1) = 1;
    u_init(1:N+1,k*N+1) = 1;

    v_init(1,1:k*N+1) = 0;
    v_init(N+1,1:k*N+1) = 0;
    v_init(1:N+1,1) = 0;
    v_init(1:N+1,k*N+1) = 0;
end

if bc == 2
    u_init(1,1:k*N+1) = 1;
    u_init(N+1,1:k*N+1) = 1;
    u_init(1:N+1,1) = 1;
    u_init(1:N+1,k*N+1) = u_init(1:N+1,k*N);
```

```
    v_init(1,1:k*N+1) = 0;
    v_init(N+1,1:k*N+1) = 0;
    v_init(1:N+1,1) = 0;
    v_init(1:N+1,k*N+1) = v_init(1:N+1,k*N);
end

u=u_init;
unew=u;
v=v_init;
vnew=v;
n = T/dt;

for t = 1:n
    rtot=1;
    rtota=[];

    vrtot=1;
    vrtota=[];

    % u velocity
    uk = u;
    while rtot >= 1e-5

        rtot = 0;
        temp=uk;
        for i = 2:N
            for j = 2:k*N
                temp(i,j)=(u(i,j) ...
                    -(u(i,j)*(u(i,j+1)-u(i,j-1)))*c ...
                    -(v(i,j)*(u(i+1,j)-u(i-1,j)))*c ...
                    + R*(temp(i-1,j)+temp(i+1,j)+temp(i,j+1)+temp(i,j-
1)))/(1+4*R);
                if bc == 2
                    temp(i,end) = temp(i,end-1);
                end
            end
        end
        uk_1 = temp;
        for i = 2:N
            for j = 2:k*N
                res=abs(uk_1(i,j)-uk(i,j));
                rtot = rtot + res;
            end
```

```
        end
        rtota=[rtota,rtot];
        uk = uk_1;
    end
    u = uk_1;

    % v velocity
    vk = v;
    while vrtot >= 1e-5

        vrtot = 0;
        vtemp = vk;
        for i = 2:N
            for j = 2:k*N
                vtemp(i,j)=(v(i,j) ...
                    -(u(i,j).*(v(i,j+1)-v(i,j-1)))*c ...
                    -(v(i,j).*(v(i+1,j)-v(i-1,j)))*c...
                    + R*(vtemp(i-1,j)+vtemp(i+1,j)+vtemp(i,j+1)+vtemp(i,j-
1)))/(1+4*R);
                if bc == 2
                    vtemp(i,end) = vtemp(i,end-1);
                end
            end
        end
        vk_1 = vtemp;
        for i = 2:N
            for j = 2:k*N
                res=abs(vk_1(i,j)-vk(i,j));
                vrtot = vrtot + res;
            end
        end
        vrtota=[vrtota,vrtot];
        vk = vk_1;
    end
    v = vk_1;

end

end

function w = vorticity(u,v,dx,k)
    N = 1/dx;
    dy = dx;
    w = zeros(N+1,k*N+1);
```

```
    for i = 2:N
        for j = 2:k*N
            dudy(i,j) = (u(i+1,j)-u(i-1,j))/2/dy;
            dvdx(i,j) = (v(i,j+1)-v(i,j-1))/2/dx;
            w(i,j) = dvdx(i,j)-dudy(i,j);
        end
    end

end
```